

ADRIAN MATIAS

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PROFILE

Data Scientist / ML Engineer with 12 years shipping end-to-end machine learning from problem framing to production. Deep roots in core DS — ad-tech ranking & user acquisition (Appodeal, Chartboost), fraud detection, NLP (RavenPack), recommender systems, Bayesian Bandits / RL, and large-scale experimentation on Spark / Databricks — with current work in GenAI: VET, PwC Advisory's LLM-driven vendor-evaluation platform (Azure OpenAI via Cavendish) — grounding raw vendor documentation into cited evidence, scoring engines, and contract-first pipelines. Held the correctness-gatekeeper seat: ~15% of the backend codebase written, half of peer PRs reviewed. Approach driven by simplicity, first-principles thinking, and measurable business outcomes.

EXPERIENCE

Senior AI Engineer

Cavendish Professionals → PwC Advisory (long-term contract) | León, Spain – Remote | *May 2026 – Present*

Senior AI Engineer on the build of VET (Vendor Evaluation Tool), an LLM-driven platform PwC Advisory uses to evaluate vendors with contractually binding outcomes: ingest heterogeneous vendor submissions and raw documentation → ground extraction in cited evidence → score → compare. Ten-week engagement, four roles held in sequence — backend feature developer, contract & schema owner, assessment-workbook lane owner, and finally correctness gatekeeper & requirements arbiter. Wrote roughly one line in seven of the codebase and reviewed half of all PRs the team opened; the role drifted from writing features to deciding what counts as correct. Stack: Python, FastAPI, Pydantic, PostgreSQL, Azure OpenAI, Next.js, GitHub Actions; deployed on internal AKS.

Contribution — 10 weeks, Jun–Aug 2026

- **Backend surface ownership** — roughly one line in seven of the production codebase, across ingestion, extraction, scoring, and aggregation; carried half of all peer review as the engagement's correctness gate.
- **Contract-first architecture** — typed contracts (Pydantic) as the single source of truth end-to-end, CI gates that fail on wire drift, versioned schema lanes for parallel change, and a local end-to-end harness that makes assessment runs reproducible outside the deployed stack.
- **Evidence grounding & provenance** — document-level provenance so every extracted value traces to cited source; validated scoring paths for weighting defects (caught and fixed a headline vendor score that ignored framework dimension weights).
- **Correctness gatekeeping at pair strength** — formal change-request authority that blocked merges until fixed; review traffic with the senior counterpart symmetric — a working pair, not a seniority gradient; load-bearing scoring and framework defects caught pre-merge.
- **Requirements arbitration** — in the engagement's second half, the arbiter of what counts as correct: raised the cross-cutting defects and requirements that others then owned to completion.
- **Enforcement that outlives the engagement** — correctness gates, machine-readable requirement contracts, and review standards now live in the repository's CI: whoever inherits it inherits the enforcement, no original author required.

What VET is

- AI-assisted vendor evaluation for contractual, binding decisions: vendor submissions and documentation grounded into cited, auditable evidence per requirement; heterogeneous data aggregated by domain experts into decision-ready scores.
- Built on Azure OpenAI with a contract-first backend (typed Pydantic contracts, CI enforcement, reproducible local assessment runs). Not yet in production — the engagement was measured on delivered correctness, not live traffic.

Data Scientist → DS Technical Lead

Appodeal: appodeal.com | León, Spain – Remote | *Dec 2024 – Jan 2026*

Developed install prediction models for mobile user acquisition: cold-start, transformers, user embeddings, transfer learning, MLOps, production pipelines, parity resolution, experimentation, and monitoring. Progressed toward technical leadership based on ownership and strategic perspective.

Key Achievements

- Unified modeling across targets, advertisers, and platforms via attention/transformer architecture, increasing end KPI.
- Enabled seamless advertiser onboarding through novel cold-start techniques (custom PyTorch + transfer learning).
- Resolved parity mismatches and calibration gaps; productionized parity workflows with unit tests and decoupling in Databricks.
- Transfer-learning user embeddings achieving performance parity with orders-of-magnitude cost savings.

Senior Data Scientist

Quant AI Lab: quant.global | León, Spain – Remote | *Jul 2024 – Dec 2024*

Designed and delivered end-to-end ML solutions for international clients; reviewed technical scope, aligned with stakeholders, and supervised model productivization.

Key Contributions

- Led rapid turnaround of underperforming client ML system (utility fraud detection & ranking): diagnosed architectural issues, executed a surgical plan, restored accuracy/efficiency, and recovered client satisfaction.
- Discovered critical issues in data ingestion and feature-engineering pipelines.
- Led real-time credit card fraud classifier combining scalable ML with deep VAE. Open version: github.com/adrianmatias/ml-sandbox/tree/master/tx_class

Senior Data Scientist

Chartboost: chartboost.com | León, Spain – Remote | *Jun 2023 – Jun 2024*

Developed ad ranking and reinforcement learning systems for in-app game advertising; advanced DSP performance through large-scale experimentation and behavioral analysis.

Key Achievements

- Evolved and monitored Bayesian Bandit RL system with policy evaluation tools; designed tailored online/offline evaluation frameworks.
- Built Two-Towers neural ranking algorithm based on recommender systems.
- Post-mortem analysis of adversarial DSP dynamics and model feedback loops.

Machine Learning Engineer

RavenPack | León, Spain – Remote | *Mar 2022 – Jan 2023*

Researched, implemented, and maintained text analytics, NLU, and semantic search products on massive financial news data; productionized prototypes with emphasis on MLOps and reproducibility.

Key Achievements

- Few-shot search personalization: simplified architecture, reduced error 30%.
- Active learning PoC for learn-to-rank: projected 300% annotation efficiency gain.
- Novel long-text fingerprinting algorithm for duplicate content detection.

Earlier Roles

Numbrs *Aug 2020 – Oct 2021*

Extended Deep Learning / Active Learning for transaction processing; scaled classifier across countries with GDPR-compliant pipelines.

Stratio BD *Aug 2019 – Aug 2020*

Propensity/churn on Spark; BERT+graph email segmentation; multi-source weather DB (Elasticsearch/Flask).

Digilant *Jun 2015 – Apr 2017*

Redesigned Consumer Persona ML model; Game Theory attribution prototype; co-directed Master's thesis.

IEBS School *Oct 2019 – Oct 2021*

Webinars and masterclasses on real-time Big Data, AI, and ML applications.

DOCOMO DIGITAL *Apr 2017 – Aug 2019*

Led fraud detection (99.8% specificity, <50 ms) and default-payment filtering; mentored juniors.

2007–2015

Civil/tunnel engineering; university teaching: Math, Physics, Chemistry.

Freelance

Freelance Data Scientist / ML Engineer | *Mar 2022 – Jun 2025*

On-demand projects in semantic search, time series forecasting, and portfolio optimization for financial analysis.

SKILLS

Core Technologies	Machine Learning & GenAI
Python, FastAPI, Pydantic, PostgreSQL, PyTorch, Transformers, Azure OpenAI / Azure / AKS, AWS/S3, Databricks, PySpark, Pandas, SQL, Docker, MLflow, Git, GitHub Actions, AI-augmented tools (Ollama, llama.cpp, Hermes Agent)	LLMs, RAG, agentic systems, LangChain / LangGraph-style orchestration, document extraction & provenance, scoring engines, Deep Learning (attention, embeddings, transfer learning), Recommender Systems, NLP, RL / Bayesian Bandits, MLOps (parity, A/B, monitoring), Few-Shot / Active Learning, Ranking
Engineering Practices	Consulting & Soft Skills
Clean code, typed contracts as source of truth, CI drift gates, OOP & FP, TDD / local e2e harnesses, reproducible pipelines, Agile / XP, production ownership	Multi-client consulting delivery, stakeholder communication, correctness gatekeeping & requirements arbitration, mentoring, business impact focus, simplification / unification

SELECTED COURSES & CERTIFICATIONS

- Claude Code: A Highly Agentic Coding Assistant: Coursera (2026)	- Functional Programming in Scala Specialization: Coursera (9.2/10, 2017)
- ML Engineering for Production (MLOps) Specialization: Coursera (2021)	- Bitcoin and Cryptocurrency Technologies: Coursera (9.5/10, 2017)
- Deep Learning with PyTorch: Siamese Networks (2022)	- Algorithms & Data Structures with Java (I, II): Coursera (8.5/10, 2014)
- Deep Learning Specialization: Coursera (2016)	- Analytics Edge: Data Science & Predictive Models: MITx (9.3/10, 2014)
- Structuring Machine Learning Projects: Coursera (2018)	- Introduction to Programming & Data Science Using Python (I, II): MITx (8.7/10, 2014)
- Apache Spark for Big Data & Scalable ML: Coursera (9.9/10, 2015)	

SELECTED PROJECTS

not the tao with adrian matias: youtube.com/@notTheTaoWithAdrianMatias Podcast exploring simplicity, consciousness, reality, machine learning, coding, sovereign AI. RAG Blog System: github.com/adrianmatias/ml-sandbox/tree/master/ragblog Retrieval-augmented generation pipeline integrating LLMs, vector embeddings, and web crawling. The Mirror: https://github.com/adrianmatias/the-mirror Knowledge repository based on llm-wiki for personalized assistant and insight for agentic AI. Credit Card Transaction Fraud Classifier: github.com/adrianmatias/ml-sandbox/tree/master/tx_class Real-time fraud detection API (classical + unsupervised signals). Deploy-ready: FastAPI, Docker, RayServe.
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EDUCATION

Master's in Geological Engineering: University of Oviedo (2010–2012) Degree in Mining & Energy Engineering: University of León (2012) Bachelor in Mining Engineering: University of León (2003–2007)

References available on request